

Methods

1. [A,B,C,D should be written in a single java file]

- A. Write a method called **evenChecker** that takes an **integer** number as its argument and prints whether the number is even or odd **inside the method**.

Sample Method Call	Sample Output
<code>evenChecker(10);</code>	Even!!
<code>evenChecker(17);</code>	Odd!!

- B. Write a method called **isEven** that takes an **integer** number as an argument and **returns** boolean true if the number is even otherwise **returns** boolean false.

Sample Method Call	Sample Output
<code>boolean result = isEven(10); System.out.println(result);</code>	true
<code>boolean result = isEven(17); System.out.println(result);</code>	false

- C. Write a method called **isPos** that takes an **integer** number as an argument and **returns** boolean true if the number is positive otherwise **returns** boolean false.

Sample Method Call	Sample Output
<code>boolean result = isPos(-5); System.out.println(result);</code>	false
<code>boolean result = isPos(12); System.out.println(result);</code>	true

- D.** Write a method called **sequence()** that takes an **integer** in its parameter called **n**. Now, if **n** is **positive** then it prints all the **even** numbers from **0** to **n**, otherwise if **n** is **negative** it prints all the **odd** numbers from **n** to **-1**.

Note: **You must call** the methods from **CW-1B** and **CW-1C**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output	Explanation
sequence(10);	0 2 4 6 8 10	Here, 10 is positive so 0,2,4,6,8,10 were printed.
sequence(-7);	-7 -5 -3 -1	Here, -7 is negative so -7,-5,-3,-1 were printed.
sequence(7);	0 2 4 6	Here, 7 is positive so 0,2,4,6 were printed
sequence(-8);	-7 -5 -3 -1	Here, -8 is negative so -7,-5,-3,-1 were printed.

2. [A,B,C should be written in a single java file]

- A.** Write a method called **circleArea** that takes an **integer** radius in its parameter and **returns** the **area** of the circle.

Note: area of a circle is πr^2

Sample Method Call	Sample Output
double area = circleArea(5); System.out.println(area);	78.5398

- B.** Write a method called **sphereVolume** that takes an **integer** radius in its parameter and **returns** the **volume** of the sphere.

Note: volume of a sphere is $\frac{4}{3}\pi r^3$

Sample Method Call	Sample Output
<pre>double volume = sphereVolume(5); System.out.println(volume);</pre>	523.5987

C. Write a method called **findSpace** that takes two values in its parameters one is an **integer** diameter and another one is a String. Using the given diameter, this method should calculate the Area of a circle or the Volume of a sphere depending on the value of the second parameter. Finally, it should print the result **inside the method**.

Note: **You must call** the method written in task **CW-2A & CW-2B**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output
<code>findSpace(10,"circle");</code>	78.5398
<code>findSpace(5,"sphere");</code>	33.5103
<code>findSpace(10,"square");</code>	"Wrong Parameter"

3. Write a method called **isPrime** which takes an integer in its parameter to check whether a number is prime or not. If the number is prime then the method returns boolean **true** otherwise it returns boolean **false**.

Sample Input	Sample Output
<pre>boolean check = isPrime(7); System.out.println(check);</pre>	true
<pre>boolean check = isPrime(15); System.out.println(check);</pre>	false

4. Write a method called **isPerfect** which takes an integer in its parameter to check whether a number is perfect or not. If the number is perfect then the method returns boolean **true** otherwise it returns boolean **false**.

Sample Input	Sample Output
<code>boolean check = isPerfect(6); System.out.println(check);</code>	true
<code>boolean check = isPerfect(33); System.out.println(check);</code>	false

5. Write a method called **calcTax** that takes 2 arguments which are **your age** then **your salary**. The method must calculate and **return** the tax as per the following conditions:

- No tax if you are less than 18 years old.
- No tax if you get paid less than 10,000
- 7% tax if you get paid between 10K and 20K
- 14% tax if you get paid more than 20K

Sample Method Call	Output	Explanation
<code>double t = calcTax(16,20000); System.out.println(t);</code>	0.0	Here, the age is less than 18 so 0 tax.
<code>double t = calcTax(20,18000); System.out.println(t);</code>	1260.0	Here, the age is greater than 18 and income is between 10K-20K so tax is 7% of 18000 = 1260.